

Zsolt Varjú

Software Developer

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Summary

Software developer and platform operator, deliberately broader than full stack. I build web applications end to end, design the ETL pipelines that feed them, and run the production infrastructure they depend on. On a project for the European Space Agency I own and run a platform that delivers quality reports and data unavailability reports from Earth observation satellites, aggregating validated data from ESA, legacy and third-party missions, from choosing the approach to keeping it running in production. AI-assisted development is a core working method, not a novelty. Heading towards system architecture.

Experience

Software Developer

Serco Czech Republic s.r.o. · Prague, Czechia · Feb 2025 – Present

- Own and run, end to end, a platform that delivers quality reports and data unavailability reports from Earth observation satellites on a project for the European Space Agency, aggregating validated data from ESA, legacy and third-party missions: from the user interface down to the servers it runs on.
- Choose the approach, technologies and architecture for new work, present the solution, build it, test it on UAT, deploy it to production, then monitor, fix, patch and back it up.
- React frontend and Node.js backend over a search stack built on Apache Solr, with RabbitMQ for messaging, Neo4j for relationships and Tika for document extraction, all containerised.
- Design and operate multi-mission ingestion pipelines and scripts in Python: scheduled transfers, search schema design, and table extraction from PDFs and spreadsheets with rejection audits returned to the data owners.
- Run Ubuntu servers on Azure: nginx routing and TLS, access control, containerised services, CI/CD, updates and patches, backups, monitoring, and regular vulnerability remediation under formal change control.
- Led a phased frontend quality programme: vendor bundle cut by roughly three quarters, JavaScript payload down a third, CSS down more than half, every outstanding lint error cleared, unnecessary React re-renders reduced by 80%.
- Work directly with mission teams and IT: requirements, brainstorming and solution proposals, formal bug reports, escalations, change records and effort estimates.

Quality Assurance Tester

Galileo Corporation s.r.o · Remote · 2022 – 2025

- Manual functional testing of web pages across browsers and devices, identifying and reporting defects.
- Verified pages against graphic design specifications and fed detailed corrections back to designers and developers.
- Maintained quality standards across a large portfolio of Slovak municipal and commercial websites.
- Documented test results and corrective actions, working independently to fixed deadlines.

Skills

Languages: TypeScript, JavaScript, Python, Bash, SQL

Web: React, Next.js, Node.js, Express, REST APIs

Search & data: Apache Solr, Open Semantic Search, Tika, Neo4j, RabbitMQ, PostgreSQL, MongoDB

ETL & automation: Custom ingestion pipelines, Cron orchestration, PDF & spreadsheet extraction, Workflow automation

AI & LLM: RAG architecture, Embeddings, Vector search (Qdrant), Hybrid retrieval, Prompt engineering, Claude Code, Codex, OpenRouter, Self-hosted models

DevOps & cloud: Docker, Kubernetes, Jenkins, GitHub Actions, Azure, AWS, Dokku, nginx, WireGuard, Tailscale

Security & practice: Vulnerability remediation, TLS, Hardening, Change management, Agile, API testing (Postman)

Projects

Self-Hosted RAG Search

Python, Qdrant, bge-m3, LangChain, Docker

Retrieval-augmented search over a local document collection: hybrid BM25 and vector retrieval, answers grounded in cited passages, and explicit abstention when no supporting evidence is found.

Coffece (coffece.sk)

Next.js, TypeScript, Tailwind CSS, next-intl

Bilingual SK/EN marketing and ordering site for an office coffee delivery brand, with localised routing, validated order forms and full SEO metadata.

IDT PLUS STAV (idtplusstav.sk)

Next.js, TypeScript, MDX, Dokku, Google Workspace

Complete digital operations for a construction SME: self-hosted marketing site, DKIM-authenticated business email, invoicing configured for Slovak VAT reverse charge, plus Google Ads and SEO.

B2B Wholesale Ordering Portal

WooCommerce, B2BKing, Stock Locations

Production B2B ordering portal for a Slovak wholesaler: tiered customer pricing, login-only catalogue, two-warehouse inventory, decimal quantities for weighted goods and integrated invoicing.

Lemma: Vocabulary Scanner

React Native, Expo, Vision models, FSRS

Mobile app that turns a photo of a printed page into vocabulary cards, pairing a vision-language detection pipeline with FSRS spaced-repetition scheduling.

Personal Infrastructure & Homelab

Linux, Docker, WireGuard, Tailscale, CI/CD

Personal VPS and home server run with production habits: WireGuard private networking, atomic zero-downtime deploys with rollback, self-healing service recovery and scheduled backups.

Education

MSc, Applied Informatics

Constantine the Philosopher University in Nitra · 2018 – 2024

Thesis: InfoMap SK, a full-stack application visualising Slovak open data from the Statistical Office API on an interactive map.

Certifications

Certified Kubernetes Administrator (CKA) · Linux Foundation / CNCF

Ubuntu Linux Professional Certificate · Canonical · Apr 2025

Learning Bash Scripting · LinkedIn Learning · Jul 2025

Basics of ABAP Programming on SAP BTP · SAP · Oct 2024

AWS Academy Graduate, Cloud Foundations · AWS Academy · Jul 2022

Languages

Hungarian (native) · Slovak (fluent) · Czech (fluent) · English (fluent)